

3	(a) displacement / velocity / acceleration / momentum / etc. three correct (none wrong) 2, two correct (none or one wrong) 1	A2	[2]
	(b) (i) $Y = 70 \text{ N}$ [allow 71 N as $+1/2$ small square on graph]	A1	[1]
	(ii) $\theta = 90^\circ$	M1	
	(for equilibrium) the direction of Y must be <u>opposite</u> to Z		
	or using $Y \sin \theta = Z$, hence $\sin \theta = 70 / 70 = 1$, $\theta = 90^\circ$	A1	[2]
	(iii) 1. $Y \cos \theta = 160$ and $Y \sin \theta = 70$	C1	
	$\tan \theta = 70 / 160$ hence $\theta = 23.6^\circ$ (24°)	A1	[2]
	2. $Y = 160 / \cos 23.6^\circ$ or $70 / \sin 23.6^\circ$ $= 174.6$ or 175 or 170 N	C1 A1	[2]
	or:		
	$160^2 + 70^2 = Y^2$	(C1)	
	$Y = 174.6$ or 175 or 170 N	(A1)	

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(c) (equilibrium not possible as) there is no vertical component from Y to balance Z B1 [1]